Answer **one** question from this Section in the spaces provided.

9 **(a)** **(i)** Explain what is meant by the *internal energy of an ideal gas*.

....

.... [1]

(ii) A sealed container of volume V contains ideal gas at pressure p.

Derive an expression for the internal energy of the gas in terms of p and V.

[2]

(b) **(i)** The mass of one mole of carbon dioxide is 44.0 g. Show that the mass of one molecule of carbon dioxide is about 7.31×10^{-26} kg.

[1]

(ii) The surface temperature of the planet Mercury is 467 °C.

Determine the root-mean-square speed of one molecule of carbon dioxide at this temperature.

root-mean-square speed = ... $m\ s^{-1}$ [3]

(iii) The escape speed of Mercury is 4250 $m\ s^{-1}$.

Suggest, with a reason, whether an atmosphere of carbon dioxide can exist on the surface of Mercury.

....

....

.... [2]

(c) (i) State the first law of thermodynamics.

....

....

.... [2]

(ii) A fixed mass of ideal gas undergoes the cycle ABCDA, as shown in Fig. 9.1.

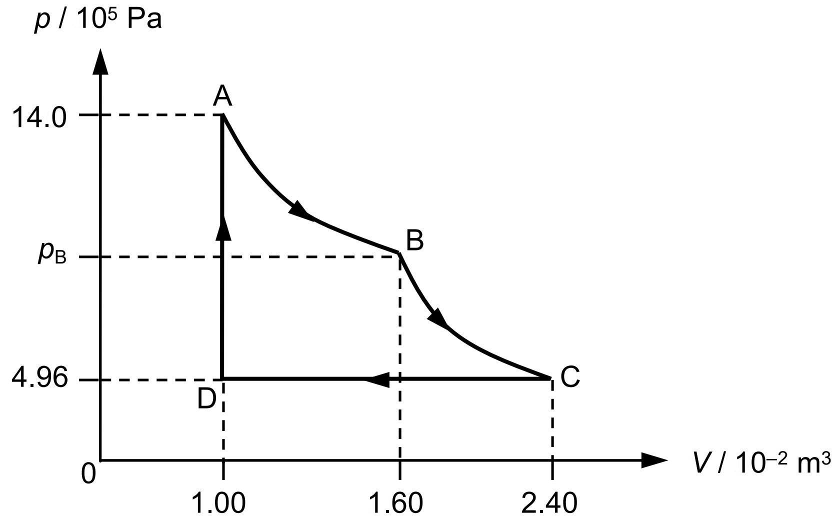

Fig. 9.1

The gas expands from A to B isothermally. It undergoes further expansion from B to C without any exchange of thermal energy.
The gas is compressed at constant pressure from C to D.
From D to A, the pressure of the gas is increased at constant volume.

1. Show that the pressure of the gas at point B is 8.75×10^5 Pa.

[1]

2. Some energy changes for the cycle in Fig. 9.1 are shown in Fig. 9.2.

	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J
A to B		– 6580	
B to C			– 3140
C to D			– 10400

Fig. 9.2

Complete Fig. 9.2. [3]

3. Calculate the increase in the internal energy of the gas from D to A.

increase = ... J [2]

(iii) The ideal gas in part **(ii)** is transferred into a container with a movable frictionless piston of small mass, as shown in the Fig. 9.3.

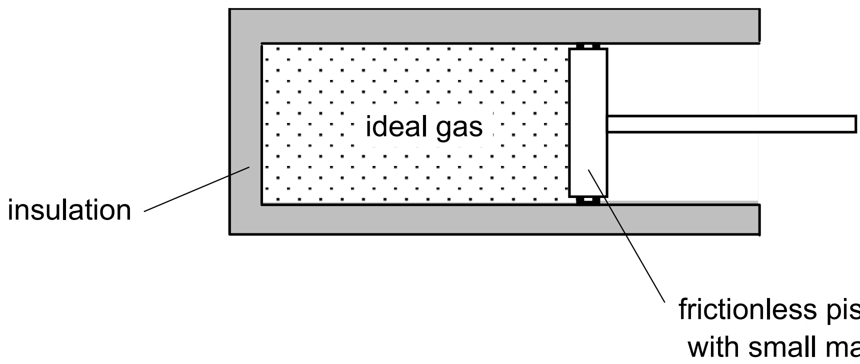

Fig. 9.3

The container and the piston are perfect thermal insulators.

Explain, in terms of the motion of the ideal gas molecules, why the temperature of the gas decreases when the gas expands.

....

....

....

....

....

.... [3]

[Total: 20]